Jacob A. Johnson

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EDUCATION

Ph.D. in Statistics Expected May 2029

University of Wisconsin-Madison, *Madison, WI*

- Advisor: Matthias Katzfuss

B.S. in Statistical Science, Minor in Mathematics June 2024

Brigham Young University, *Provo, UT*

- *summa cum laude* (4.00/4.00 GPA)
- National Merit Scholar

RESEARCH & PROFESSIONAL EXPERIENCE

Graduate Statistics Intern May 2025 – Present

Sandia National Laboratories, *Albuquerque, NM*

- Developed statistical machine learning models to forecast extreme temperatures at subseasonal (2–8 weeks out) timescales across the contiguous United States.
- Evaluated and developed various dimension reduction techniques to extract predictive features from 40+ years of atmospheric data to be used as inputs for temperature forecasting.

Financial Data Analytics Intern May 2024 – Aug 2024

Williams International, *Pontiac, MI*

- Created statistical models to forecast inventory using historical production and usage data.
- Developed an anomaly detection program to monitor and identify deviations in expenses, detecting unusual spending patterns and potential fraud across business units.
- Automated forecasting and anomaly detection processes, reducing manual oversight and providing executives with actionable insights.

Research Assistant May 2022 – April 2024

Brigham Young University, *Provo, UT*

- Utilized Bayesian machine learning techniques to combine climate data products for predicting average monthly precipitation in the Himalayan Mountains.
- Developed new subsampling methods in R to speed up computation of Gaussian process calculations for large spatial data.

PUBLICATIONS

[1] **Jacob A. Johnson**, Matthew J. Heaton, William F. Christensen, Lynsie R. Warr, and Summer B. Rupper. "Fusing Climate Data Products Using a Spatially Varying Autoencoder". In: *Journal of Agricultural, Biological and Environmental Statistics* 31.2 (June 2026), pp. 273–286. DOI: 10.1007/s13253-024-00657-3.

[2] Matthew J. Heaton and **Jacob A. Johnson**. "Minibatch Markov Chain Monte Carlo Algorithms for Fitting Gaussian Processes". In: *Bayesian Analysis* 21.1 (Mar. 2026), pp. 21–43. DOI: 10.1214/24-BA1459.

AWARDS & HONORS

- UW-Madison Statistics Department Outstanding Performance as a TA award 2024 – 2025
- ASA Section on Statistics and the Environment Student Paper Competition Winner 2024

PRESENTATIONS

Invited Presentations

- Climate Data Melding via Spatially Varying Bayesian Autoencoder
  Joint Statistical Meetings (JSM), Portland, OR Aug 2024

Contributed Presentations

- A Variational Ensemble Transport Filter for Nonlinear Data Assimilation
  JPL Uncertainty Quantification for Remote Sensing Virtual Meeting Oct 2026
- Enhancing Subseasonal Temperature Forecasts via Machine Learning and Nonlinear Dimension Reduction
  Joint Statistical Meetings (JSM), Boston, MA Aug 2026
- Modeling Bayesian Transport Map Uncertainty for Non-Gaussian Spatial Data via Laplace Approximation
  Joint Statistical Meetings (JSM), Nashville, TN Aug 2025
- Minibatch Markov Chain Monte Carlo Algorithms for Fitting Gaussian Processes
  CPMS Student Research Conference, Provo, UT Feb 2024
- Climate Data Melding via Spatially varying Bayesian Autoencoder
  Joint Statistical Meetings (JSM), Toronto, ON Aug 2023
  CPMS Student Research Conference, Provo, UT Feb 2023

Poster Presentations

- A Variational Ensemble Transport Filter for Nonlinear Data Assimilation
  ENVR Workshop, Houston, TX Oct 2026
  Fall CIRCS IAB Meeting, Madison, WI Oct 2026
- Minibatch Markov Chain Monte Carlo Algorithms for Fitting Gaussian Processes
  ENVR Workshop, Boulder, CO Oct 2024
- Climate Data Melding via Spatially Varying Bayesian Autoencoder
  ENVR Workshop, Provo, UT Oct 2022

TEACHING EXPERIENCE

Teaching Assistant, *University of Wisconsin-Madison* Sept 2024 – April 2026

- STAT 240 – Data Science Modeling I (Fall 2025, Spring 2026)
- STAT 324 – Introductory Statistics for Engineers (Fall 2024, Spring 2025)

Teaching Assistant, *Brigham Young University* Sept 2021 – April 2022

- STAT 240 – Probability and Inference I (Fall 2021, Spring 2022)

SKILLS

- **Technical:** R, Python, SQL, Linux/shell scripting.
- **Languages:** English (Native), Spanish (Fluent).